

Matt J Bryan

BA MENG (CANTAB) MIET

PhD Student,  Cambridge University Engineering Department (CUED)

Cambridge, UK |  mjb314@cam.ac.uk |  [mjb314.github.io](https://github.com/mjb314) |  | 

Education

UNIVERSITY OF CAMBRIDGE - MAGDALENE COLLEGE

OCT 2024 –

→ PhD, ENGINEERING - fully funded by Bose (MA, USA)

- **Project:** *Hybrid physics-ML models for active noise control in automotive systems*
- **Research:** Incorporating underlying physics into machine learning (ML) architectures to improve performance in active noise control (ANC), which requires models to be *both* accurate and fast.
- This combines expertise in physical domains (nonlinear dynamics, vibration, acoustics) with control theory and data-driven ML methods to solve complex problems e.g. real-time prediction of sound through a vehicle.
- Tailored physics-ML architectures show benefits in nonlinear time-series prediction. Python/Torch used.
- **Supervisors:** Dr Tore Butlin (CUED) and Dr Ole Nielsen (Bose/CUED).

UNIVERSITY OF CAMBRIDGE - MAGDALENE COLLEGE

2020 – 2024

→ BA MENG, ENGINEERING (MECHANICAL & CONTROL/ML)

- **Grades:** 1st Class, top 10% (BA), Distinction, top 15% (MEng). 1st every year.
- Two general years, before dual specialisation in Mechanical and Control/ML Engineering in the third year. Advanced study of vibration, nonlinear dynamics, control, and acoustics in the fourth year.
- **MEng Project:** ‘Pushing the bounds of energy harvesting’ - 1st Class.
Investigated circumventing the ‘mass bound’ for general energy harvesting systems. Developed a design from theory to prototype with a novel dual translation and rotationally excited piezoelectric harvester.

Teaching

Supervisor - MAGDALENE/CLARE/JESUS COLLEGE, CAMBRIDGE

OCT 2024 –

- Passionate about teaching across all levels of engineering with c. 30 students p.a. In high demand as a supervisor with strong results - my first year students scored 10% higher than the cohort average.
- Supervising students for 1st year Mechanics at Magdalene, 2nd year at Clare and Jesus, and 3rd year advanced Dynamics and Vibration courses. Plus, a number of ‘Bicycle Design’ research projects.
- Completed an **AFHEA** teaching qualification through selective University Scheme.

Admissions Interviewer - MAGDALENE/EMMANUEL COLLEGE, CAMBRIDGE

DEC 2024 –

- One of Magdalene’s youngest ever interviewers (at 22 years-old). Design of technical questions, conducted 80+ engineering undergraduate admission interviews. Contributing to decision and pooling process.
- Working closely with Directors of Studies and Admissions Tutors. Also invited to interview for Emmanuel.

Industrial Experience

Research Intern - BOSE RESEARCH, FRAMINGHAM MA

JUN - AUG 2026

- Research visit to Bose, working with the research team in Framingham MA, USA. Working on physics-informed machine learning on a full scale nonlinear automotive system as part of my PhD.

ML Engineer - ENTANGL, SAN FRANCISCO CA

APR - MAY 2026

- Part-time remote machine learning engineer for a start-up developing models for data-centres.

- Two summers working with superconductive MRI systems - novel vibration measurement technique in a strong magnetic field and finite-element simulation of undesirable field-induced vibration to inform design.
- Highly competitive selection for Siemens' Cambridge Sponsorship Scheme (1 of 2 across Cambridge).

Publications

[1] PATENT - **Novel machine learning architecture for active noise control in nonlinear systems**

- From my PhD work - application filed with Cambridge Enterprise and Bose in May 2026.

[2] PAPER - **Physics-informed machine learning models for automotive active noise control**

- Accepted conference paper, to be presented at ISMA26, Leuven BE, September 2026.

Academic Activities

- **Convener:** Dynamics and Vibration Research Group (DVRG) Seminar Series, 2025/26
Scheduling and chairing weekly talks with internal and invited speakers across dynamics, vibration, and ML.
- **Talks Given:**
 - '*Hybrid physics-ML models for automotive ANC*' - Bose Research Series, Framingham MA, Aug 2026.
 - '*Hybrid physics-ML models for automotive ANC*' - Tea Time Talk, Cambridge, Jun 2026.
 - '*Hybrid models for Active Noise Reduction*' - Institute of Physics PIML Workshop, London, Oct 2025.

Awards

Best Poster - CUED Division C Graduate Conference 2025

MAY 2025

PhD Studentship - Bose at the University of Cambridge

OCT 2024 -

- Full studentship (c.£300k) fully funded by Bose Corporation for 4 years of doctoral study, including research budget and collaboration in the US. Awarded following a competitive interview process.

College Scholarships & Prizes - Magdalene College, Cambridge

OCT 2020 -

- Current **Senior Scholar** for a Distinction in my Master's. Awarded an academic scholarship every year of my undergraduate degree for First Class results, including Christopherson and Foo Sun Lau prizes.

Undergraduate Scholarship - Institute of Mechanical Engineers

OCT 2020 - JUN 2024

- Academic scholarship awarded for my MEng following a competitive application process (1 of 6 nationally).

Extracurricular

Memberships: Institution of Engineering and Technology (MIET), Cambridge Philosophical Soc (FCPS).

Volunteering: Cambridge University Cycling Club Committee, various roles inc. General Secretary, Welfare Officer (2021 -), CUED Outreach, Cambridge Slipstream Mentoring Scheme.

Hobbies: Cycling (road and mountain-biking) → 2× Half Blue. Choral singing → Magdalene (*Choral Scholar*, Garrett Music Prize, Music Soc Treasurer) & Queens' College Choirs, Portsmouth Cathedral Choir.

Referees

Dr Tore Butlin — tb267-at-cam.ac.uk

- University Associate Professor, CUED (Principal PhD Supervisor)

Dr Richard Roebuck — r1r20-at-cam.ac.uk

- Director of Studies in Engineering (Part I), Magdalene College, Cambridge